Physical Activity Towards Health and Fitness 2

(PATH-FIT 2)

Exercise-Based Fitness Activities

Glenn Kenneth Z. Valerozo

LAGUNA UNIVERSITY

Vision

Laguna University shall be a socially responsive educational institution of choice providing holistically developed individuals in the Asia-Pacific Region.

Mission

Laguna University is committed to produce academically prepared and technically skilled individuals who are socially and morally upright.

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MODULE 4
Health Appraisal

Introduction

A health appraisal is a systematic process of evaluating an individual's health status, often conducted by healthcare professionals, wellness coaches, or individuals themselves. The purpose of a health appraisal is to gather information about various aspects of a person's health and lifestyle, helping to identify potential risk factors, establish a baseline for health measurements, and guide health-related decision-making. Health appraisals can be used in various settings, including healthcare, workplace wellness programs, and fitness assessments.

Health appraisals can take various forms, ranging from self-assessment questionnaires to comprehensive evaluations conducted by healthcare professionals. They are often used as a foundation for developing personalized health and wellness plans, guiding individuals toward healthier lifestyles, and preventing or managing health issues.

Intended Learning Outcomes

At the end of the module, the students should be able to:

1. Identify the different types of injury and coping methods.
2. Acknowledge the importance of health appraisal before, during and after Physical Activity Performance.
3. Comprehend the safety guidelines and health protocol set by the course.

Lesson 1. Safety Tips and Protocols

Exercise is a crucial component of a healthy lifestyle, but it is essential to approach it with caution and follow proper protocols to ensure safety and effectiveness. This essay will discuss exercise precautions, tips, and protocols to guide individuals in achieving their fitness goals while minimizing the risk of injuries.

Advice and instructions for teachers' practices are based on identified hazards (Podstawski et al., 2015) or scenarios that may evolve in PE classes (Merrie et al., 2016). To prevent student injuries and protect teachers from liability, an additional strand of RSM instructions stems from the analysis of tort law and teaching negligence cases (Murphy and Beh, 2014; Gimbert and Sawyer, 2015; McCoy et al., 2017).

1. **Pre-Exercise Precautions:**
    - **Health Screening**: Individuals should undergo a health screening or seek medical clearance before starting a new exercise program, especially if they have existing health conditions.
    - **Risk Assessment:** Self-assessment of personal health risks and factors that may impact the ability to engage in certain exercises.
    - **Consultation with Professionals:** Seek guidance from healthcare professionals or fitness experts, especially for those with specific health concerns or beginners.
2. **Safety Variables:**
    - Safety variables in exercise refer to factors and considerations that influence the safety of individuals engaging in physical activity.
    - Individuals should consider their current health status, including any pre-existing medical conditions or recent injuries, before engaging in exercise. Consulting with a healthcare professional may be necessary, especially for those with specific health concerns.
    - The fitness level of an individual plays a significant role in determining the appropriate intensity, duration, and type of exercise. Gradual progression is essential to prevent overexertion.
    - Including a proper warm-up before exercise and a cool-down afterward helps prepare the body for activity and facilitates recovery, reducing the risk of muscle strains and injuries.

- Performing exercises with proper form and technique is crucial for preventing injuries. Incorrect form can lead to strains, sprains, and other musculoskeletal issues.
- Ensuring the safety of exercise equipment is essential. Regular inspection, maintenance, and proper usage of equipment help prevent accidents and injuries.

3. Proper Clothing

- Choose clothing made of moisture-wicking materials, such as synthetic blends or specialized performance fabrics. These materials help wick sweat away from the body, keeping you dry and comfortable during exercise.
- Opt for breathable fabrics that allow air circulation to help regulate body temperature. This is especially important during intense or prolonged exercise sessions.
- Wear clothing that fits well and allows for a full range of motion. Avoid overly tight or restrictive attire that may impede movement, and steer clear of excessively loose clothing that could get caught in equipment.
- Choose athletic shoes designed for the specific activity you are engaging in. Proper footwear provides support, stability, and cushioning, reducing the risk of foot and ankle injuries.
- If exercising in low-light conditions, wear clothing with reflective elements or add reflective accessories to enhance visibility and safety, particularly when running or cycling.
- For specific activities such as cycling, running, or weightlifting, consider specialized gear designed for those activities. This includes items like padded cycling shorts, moisture-wicking running shirts, or weightlifting gloves.
- Compression garments can help improve circulation, reduce muscle soreness, and provide support during intense workouts. These items are particularly popular in activities like running, cycling, and weightlifting.
- Depending on the type of exercise, consider accessories like a supportive sports bra for women, knee braces, or ankle supports to help prevent injuries and provide added comfort.
- Keep your exercise clothing clean and well-maintained to ensure hygiene and prevent skin irritations. Replace worn-out or damaged items as needed.

4. Fluid Intake and Replacement

- Fluid intake and replacement are critical aspects of maintaining proper hydration, which is essential for overall health and well-being. Adequate hydration is

particularly important during physical activity, in hot or humid conditions, and for individuals with specific health concerns.

- Aim to consume an adequate amount of fluids throughout the day, even when not engaged in physical activity. The National Academies of Sciences, Engineering, and Medicine suggests a daily fluid intake of about 3.7 liters (125 ounces) for men and 2.7 liters (91 ounces) for women, including fluids from all beverages and foods.
- Before Physical Activity: Drink water throughout the day leading up to exercise. Consume about 16-20 ounces (500-600 ml) of water 2-3 hours before exercise, and an additional 8-10 ounces (250-300 ml) 20-30 minutes before starting.
- Drink fluids regularly during exercise, especially for activities lasting longer than 60 minutes. Aim for 7-10 ounces (200-300 ml) every 10-20 minutes.
- Individual fluid needs vary based on factors such as body weight, age, gender, and overall health. Tailor fluid intake to your specific circumstances.
- Pay attention to thirst cues. While it's a good indicator of hydration needs, it's also important to drink fluids regularly and not wait until feeling excessively thirsty.
- Adjust fluid intake based on environmental conditions. Hot and humid conditions may increase fluid requirements.
- Water is generally the best choice for maintaining hydration. It is calorie-free and effective for most physical activities.
- Consider sports drinks for activities lasting more than 60-90 minutes, as they provide electrolytes and carbohydrates for energy.
- Limit the consumption of caffeinated and alcoholic beverages, as they can contribute to dehydration.
- Consume water-rich foods like fruits and vegetables, which contribute to overall hydration.

5. Blister

- Assess the nature of activities: Certain activities, such as prolonged walking, running, or use of ill-fitting footwear, may increase the risk of blisters.
- Proper footwear: Ensure that individuals have appropriate footwear that fits well and does not cause friction.
- Protective gear: In certain workplaces or activities, protective gear like gloves can prevent blisters.
- Breaks and hydration: Encourage breaks during activities to reduce continuous friction and promote hydration to maintain skin elasticity.

- Train individuals on recognizing early signs of friction and how to address it promptly.
- Provide information on proper blister care, including avoiding popping blisters and keeping them clean to prevent infection.
- Train personnel on proper blister care procedures, including when to seek professional medical attention.
- Establish a system for monitoring and reporting incidents related to blisters.
- Ensure that individuals know when it is appropriate to seek medical assistance for severe or infected blisters.

6. Muscle Soreness

- Muscle soreness is a common occurrence, especially after engaging in physical activities or exercise.
- Always start with a proper warm-up before engaging in strenuous activities. This helps prepare the muscles for the upcoming workload. Before Physical Activity: Drink water throughout the day leading up to exercise. Consume about 16-20 ounces (500-600 ml) of water 2-3 hours before exercise, and an additional 8-10 ounces (250-300 ml) 20-30 minutes before starting.
- Include a cool-down session after exercising to gradually reduce the intensity of your workout, promoting better recovery. Individual fluid needs vary based on factors such as body weight, age, gender, and overall health. Tailor fluid intake to your specific circumstances.
- Avoid overexertion, which can contribute to muscle soreness and the risk of injury.
- Stay well-hydrated before, during, and after exercise. Dehydration can exacerbate muscle soreness.
- Ensure a balanced diet with an adequate intake of protein, carbohydrates, and essential nutrients to support muscle recovery. Limit the consumption of caffeinated
- Incorporate stretching exercises into your routine to improve flexibility and reduce muscle tension.
- Allow sufficient time for rest and recovery between intense workouts.
- Aim for 7-9 hours of quality sleep each night to support overall recovery.
- Use proper techniques to avoid causing additional strain.
- Nonsteroidal anti-inflammatory drugs (NSAIDs) may provide relief for muscle soreness, but it's advisable to consult with a healthcare professional before use.

- Ensure you have proper footwear and equipment for your activities to reduce the risk of muscle strain.
- Pay attention to your body's signals. If you experience persistent or severe muscle soreness, it may be a sign of overtraining or an underlying issue.
- Consult with a fitness professional or healthcare provider to ensure your exercise routine is appropriate for your fitness level and health condition.
- Ice packs can help reduce inflammation, while heat therapy can improve blood flow and promote muscle relaxation. Use them as appropriate for your situation.

7. Sprain and Strains

- Strains and sprains are common injuries that can occur when muscles or ligaments are stretched or torn.
- Always engage in a proper warm-up before physical activities or exercise to prepare muscles and ligaments for the workload.
- Incorporate strength training exercises to build muscle strength, which can help stabilize joints and reduce the risk of strains and sprains.
- Learn and use proper techniques for exercise and physical activities to avoid putting unnecessary stress on muscles and ligaments.
- Include regular flexibility exercises to maintain joint range of motion and prevent stiffness.
- Wear proper footwear for different activities to provide support and reduce the risk of ankle and foot sprains.
- Stay well-hydrated to support the elasticity of muscles and ligaments.
- Allow sufficient time for rest and recovery between intense workouts to prevent overuse injuries.
- Pay attention to your posture, especially during activities that involve repetitive movements, to avoid putting excess strain on certain areas.
- Follow the R.I.C.E. protocol for immediate care: Rest, Ice, Compression, and Elevation. Rest the affected area, apply ice to reduce swelling, use compression to support the injured area, and elevate it to minimize swelling.
- Refrain from activities that worsen the strain or sprain, allowing time for the injury to heal.
- Follow any recommended physical therapy exercises or rehabilitation programs to aid recovery and prevent future injuries.
- Depending on the severity, a healthcare professional may recommend braces or

supports to stabilize the injured area during recovery.

- Gradually reintroduce activities and exercises as directed by a healthcare professional to avoid re-injury.

8. Cramps, Dislocation, and Fracture

- A dislocation occurs when the bones that form a joint are forced out of their normal position. This can happen due to trauma, injury, or a forceful impact. Common dislocations include those of the shoulder, finger, and knee.
- Cramps are sudden, involuntary muscle contractions that can be painful. They can occur for various reasons, including dehydration, electrolyte imbalances, muscle fatigue, or overuse.
- A fracture is a break or crack in a bone. Fractures can result from trauma, falls, or repetitive stress on a bone. Common types include open (compound) fractures, closed fractures, and stress fractures.
- Keep the injured joint as still as possible to prevent further damage.
- Avoid attempting to relocate the joint unless you are a trained professional.
- Apply ice to the affected area to reduce swelling.
- Elevate the injured limb if possible to further minimize swelling.
- Do not try to pop the joint back in place on your own.
- Gently stretch and massage the cramped muscle to help relieve the spasm.
- Allow the affected muscle to rest and recover.
- Avoid overexertion during physical activities.
- Immobilize the affected limb to prevent further injury. Use splints, slings, or other available materials to stabilize the fracture site.
- Do not attempt to realign the bones or manipulate the fracture unless you are a trained professional.
- Keep the injured person calm and reassured while awaiting professional medical help.

In all cases, these tips are general guidelines. For specific situations, it's essential to seek professional medical advice promptly. If you are unsure about how to handle a particular injury, it is safer to wait for medical professionals to assist. If you suspect a fracture or dislocation, avoid unnecessary movement of the affected area until professional medical evaluation occurs.

Tips on how we can minimize possible injuries:

- ✓ *Warm – Up and Stretching*
- ✓ *Proper Techniques*
- ✓ *Gradual Progression*
- ✓ *Strength Training*
- ✓ *Proper Equipment*
- ✓ *Hydration*
- ✓ *Rest and Recovery*
- ✓ *Cross-Training*
- ✓ *Environmental Awareness*
- ✓ *Listen to your body*
- ✓ *Regular Check-ups*
- ✓ *Education about safety guidelines*
- ✓ *Emergency Preparedness*
- ✓ *Professional Guidance*

Remember that individual needs and considerations may vary, so it's advisable to consult with healthcare or fitness professionals for personalized advice based on your specific health status and goals.

Assessment Task 5

The goal of this project is to assess students' comprehension and application of safety tips and protocols in various physical education activities.

Safety Poster Creation

1. **Choose an Activity**: Select a physical education activity (e.g., team sports, gymnastics, running) that interests you.
2. **Identify the Safety Tips**: Research and identify at least 10 safety tips specific to the chosen activity and tips should cover aspects such as warm-up, equipment uses, proper technique, and emergency procedures.
3. **Create a Safety Poster:** Design an informative and visually appealing safety poster using a digital tool or by hand. Include illustrations, icons, or images to enhance understanding. Clearly present each safety tip along with a brief explanation.

Safety Drill Demonstration (Group)

1. ***Form Groups**: I*n groups of 3-4, select a physical education activity different from your individual choice.
2. ***Develop a Safety Drill**:* Design a safety drill or routine related to the chosen activity. Clearly outline steps for warm-up, safe execution of the activity, and cool-down.
3. ***Demonstration**:* Perform the safety drill in front of the class, emphasizing key safety protocols. Briefly explain each step, highlighting safety measures and their importance.

Summary

Health appraisal in physical education serves as a comprehensive assessment of an individual's overall health and fitness, providing valuable insights to tailor physical activities

and promote well-being. Key components of health appraisal include physical assessments, lifestyle evaluations, and consideration of any pre-existing health conditions.

In conclusion, health appraisal in physical education is a holistic approach that goes beyond physical fitness assessments. It considers lifestyle factors, medical history, and psychosocial elements to create a comprehensive understanding of an individual's health. This information guides the development of tailored fitness plans, promotes positive lifestyle changes, and contributes to lifelong well-being.

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MODULE 5
High Intensity Interval Training

Introduction

High-Intensity Interval Training (HIIT) has revolutionized the fitness landscape, emerging as a dynamic and efficient approach to achieving optimal cardiovascular health and physical fitness. This innovative training methodology captivates fitness enthusiasts and professionals alike with its ability to deliver remarkable results in shorter durations. HIIT involves alternating short bursts of intense exercise with brief periods of rest or lower-intensity activity, challenging the body to push its limits and reap numerous health benefits.

The rise of HIIT's popularity is rooted in its versatility, time efficiency, and adaptability to various fitness levels and preferences. Whether through sprinting, cycling, bodyweight exercises, or a combination of modalities, HIIT provides a customizable and accessible avenue for individuals to elevate their fitness routines

Intended Learning Outcomes

At the end of this module the students should be able to:

1. Recognize the planes of motion and mechanics of an exercise.
2. Describe the health advantages of aerobic exercise, flexibility exercises, muscular strength exercises, and power exercises.
3. Create a customized program using interval timing techniques
4. Calculate your goal heart rate by measuring your resting heart rate.
5. Explain the rationale for altering behavior and maintaining a physical activity regimen.

Lesson 1. Dynamic Stretching

HIIT (High Intensity Interval Training)

High-Intensity Interval Training, commonly known as HIIT, is a fitness approach characterized by short bursts of intense exercise alternated with brief periods of rest or lower-intensity activity. This training methodology has gained widespread popularity due to its efficiency in promoting cardiovascular fitness, calorie burn, and various health benefits.

Key Elements of HIIT:

1. **Intensity:** The hallmark of HIIT is its high-intensity intervals, where individuals exert themselves close to their maximum effort during short bursts of activity. This elevates heart rate and challenges the cardiovascular system.
2. **Interval Structure:** HIIT alternates between work intervals and rest or active recovery intervals. Work intervals typically range from 20 to 60 seconds, followed by shorter rest periods. This cycle is repeated throughout the workout.
3. **Versatility:** HIIT workouts can be adapted to various exercises, including running, cycling, bodyweight exercises, and more. This versatility allows individuals to customize their routines based on personal preferences and fitness levels.
4. **Time Efficiency:** One of the key advantages of HIIT is its time efficiency. Despite shorter workout durations, HIIT can deliver comparable or even superior fitness benefits compared to longer, moderate-intensity workouts.

Benefits of HIIT

- ✓ Caloric Burns
- ✓ Cardiovascular Health (Improves anaerobic and aerobic capacity)
- ✓ Help to lose weight quickly
- ✓ Build muscle
- ✓ Time saving

Persuade (someone) to do HIIT:

- ✓ Muscle loss
- ✓ Muscle cramps
- ✓ Slow recovery

- ✓ Loss of energy
- ✓ Fatigue

HIIT workout should be done up **3 times/week** on alternate day and focus on healthy diet to get quick and better results.

Dynamic Stretching

Dynamic stretching is a form of stretching that involves controlled, fluid movements through a full range of motion. Unlike static stretching, where you hold a stretch for an extended period, dynamic stretching is characterized by continuous and active movements that mimic the activity or sport you're about to engage in. This type of stretching is typically done as part of a warm-up routine before exercise, sports, or any activity that requires flexibility and muscle engagement.

Key characteristics of dynamic stretching include:

1. **Continuous Movement:** Dynamic stretches involve flowing movements rather than holding a static position. This helps increase blood flow, warm up the muscles, and improve flexibility.
2. **Specific to Activity:** Dynamic stretches are often tailored to the specific movements or muscle groups that will be used in the upcoming activity. This helps prepare the body for the demands of the exercise or sport.
3. **Full Range of Motion:** Dynamic stretches are designed to take the joints and muscles through their complete range of motion. This helps improve flexibility and mobility.
4. **Gradual Intensity:** The intensity of dynamic stretching can be gradually increased as the warm-up progresses. This allows the body to adjust to the increasing demands placed on the muscles.
5. **Enhance Blood Flow:** Dynamic stretching promotes increased blood circulation to the muscles, which brings oxygen and nutrients, preparing them for more strenuous activity. Examples of dynamic stretches include:
    - *Leg swings*: Forward and sideways leg swings help warm up the hip flexors, quadriceps, and hamstrings.
    - *Arm Circles*: Circular movements of the arms in both forward and backward directions help warm up the shoulders and improve flexibility.
    - *High Knees*: Lifting the knees toward the chest in a jogging motion warms up the hip flexors and activates the core muscles.

- *Jumping Jacks*: This classic exercise engages multiple muscle groups, promoting overall warm-up.
- *Walking Lunges*: Dynamic lunges help stretch the hip flexors, quadriceps, and hamstrings while engaging the core.
- *Hip Circles*: Rotational movements of the hips in both clockwise and counterclockwise directions improve hip flexibility.

Benefits of dynamic stretching includes:

- *Improved Flexibility:* Dynamic stretching can enhance flexibility by actively engaging muscles through their full range of motion.
- *Enhanced Performance:* Dynamic stretching can enhance flexibility by actively engaging muscles through their full range of motion.
- *Injury Prevention:* Proper warm-up through dynamic stretching can help reduce the risk of injury by preparing the body for physical exertion.
- *Increase Blood Circulation:* The continuous movement in dynamic stretching promotes blood flow, delivering nutrients and oxygen to the muscles.

Lesson 2. Proper Execution of Exercise

Proper execution of exercises in physical education is crucial to ensure effectiveness, maximize benefits, and minimize the risk of injuries. Whether in a school setting or any fitness program, here are key guidelines for the proper execution of exercises: The secondary objective of biomechanics of sport and physical exercise is to make recommendations for injury prevention and rehabilitation. This secondary objective of sport biomechanics is inextricably linked to the first objective since a healthy athlete will outperform an injured athlete.

1. **Warm-Up:**
   - **Purpose:** To increase blood flow, raise body temperature, and prepare muscles for more intense activity.
   - **Execution:**
     - Include light aerobic exercises (e.g., jogging, jumping jacks).
     - Incorporate dynamic stretches targeting major muscle groups.

2. Proper Form:

- **Purpose:** Prevents injuries and ensures the targeted muscles are engaged.
- **Execution:**
    - Follow the correct body alignment for each exercise.
    - Maintain a neutral spine and proper posture.
    - Avoid using momentum to complete movements.

3. Progress Gradually:

- **Purpose:** Allows the body to adapt to increasing intensity.
- **Execution:**
    - Start with exercises that match the fitness level of participants.
    - Gradually increase intensity or resistance as strength and endurance improve.

4. Individualization:

- **Purpose:** Accommodates different fitness levels and abilities.
- **Execution:**
    - Provide variations or modifications for exercises.
    - Consider individual needs and adjust the workout accordingly.

5. Breathing Technique:

- **Purpose:** Enhances oxygen intake and supports muscle function.
- **Execution:**
    - Inhale during the easier phase of the exercise.
    - Exhale during the more challenging phase.

6. Proper Equipment Use:

- **Purpose:** Ensures safety and maximizes the effectiveness of exercises.
- **Execution:**
    - Instruct participants on the correct use of equipment (weights, resistance bands, etc.).
    - Check equipment for safety before use.

7. Cool Down:

- **Purpose:** Facilitates recovery and prevents muscle stiffness.
- **Execution:**
    - Include static stretching targeting major muscle groups.
    - Gradually decrease the intensity of activity.

8. Safety First:

- **Purpose:** Minimizes the risk of injuries.
- **Execution:**
    - Emphasize the importance of following safety guidelines.

    - Supervise exercises, especially those involving equipment or complex movements.

9. Clear Instruction:

- **Purpose:** Ensures participants understand how to perform exercises correctly.
- **Execution:**
    - Provide clear, step-by-step instructions.
    - Demonstrate exercises and correct techniques.

10. Variety and Engagement:

- **Purpose:** Keeps participants motivated and prevents monotony.
- **Execution:**
    - Introduce a variety of exercises to target different muscle groups.
    - Incorporate enjoyable and challenging activities.

11. Regular Monitoring:

- **Purpose:** Allows for adjustments and ensures continued progress.
- **Execution:**
    - Monitor participants' form and provide feedback.
    - Assess progress and adjust workout plans accordingly.

12. Encourage Communication:

- **Purpose:** Builds a supportive and open environment.
- **Execution:**
    - Encourage participants to communicate any discomfort or concerns.
    - Foster a positive atmosphere for learning and improvement.

By adhering to these guidelines, physical education instructors can promote safe and effective exercise execution, creating an environment that encourages physical fitness, skill development, and overall well-being.

Biomechanics is also applied in the design of sports equipment. This includes anything from shoes and clothing to sports equipment like bats and balls. The purpose of biomechanics in this situation is to design equipment that can improve athlete performance while reducing the risk of injury (Sri Chaitanya College, 2022)

Lesson 3. Work Out vs Rest Interval

The choice of workout and rest intervals plays a crucial role in designing an effective and goal-specific training program. The duration and intensity of these

intervals can significantly impact the physiological responses and adaptations of the body. Here's an overview of the concepts of workout and rest intervals:

Workout Interval:

Definition: The workout interval, also known as the work interval or exercise interval, is the period during which an individual engages in intense physical activity or exercises.

Key Points:

1. **Intensity:** The workout interval involves high-intensity effort where the individual exerts maximal or near-maximal effort.
2. **Purpose:** It aims to challenge the cardiovascular system, build endurance, and improve specific fitness components such as strength, speed, or power.
3. **Duration:** The duration of the workout interval can vary based on the training goal and the specific exercise being performed. It may range from seconds to several minutes.

Rest Interval:

Definition: The rest interval is the period of recovery or rest between consecutive workout intervals.

Key Points:

1. **Recovery:** The rest interval allows the body to recover partially or fully from the physiological stress induced during the workout interval.
2. **Purpose:** The primary purpose is to enable the individual to maintain the quality of effort during subsequent workout intervals, preventing premature fatigue.
3. **Duration:** The duration of the rest interval depends on factors such as the intensity and duration of the workout interval, fitness level, and training goals. It can range from a few seconds to several minutes.

Considerations for Adjusting Intervals:

1. **Training Goal:**
    - For cardiovascular endurance and fat loss, shorter rest intervals and longer workout intervals may be employed.
    - For strength and power development, longer rest intervals and shorter, high-intensity workout intervals may be used.

2. **Fitness Level:**
    - Beginners may require longer rest intervals to recover adequately.
    - Advanced individuals may tolerate shorter rest intervals as their conditioning improves.
3. **Specific Exercise:**
    - The type of exercise influences the choice of intervals. Compound movements or high-intensity exercises may require longer rest intervals.
4. **Individual Preferences:**
    - Preferences and adherence are crucial. If an individual enjoys a particular interval structure, they are more likely to stick to their workout routine.

Examples:

1. **High-Intensity Interval Training (HIIT):**
    - **Workout Interval:** 30 seconds of intense exercises (e.g., sprinting, burpees).
    - **Rest Interval:** 15 seconds of rest or active recovery (e.g., light jogging).
2. **Strength Training:**
    - **Workout Interval:** 60 seconds of weightlifting (e.g., squats).
    - **Rest Interval:** 90 seconds of rest between sets.

Conclusion:

The optimal balance between workout and rest intervals depends on individual goals, fitness levels, and the specific demands of the training program. Experimenting with different interval structures and adjusting based on individual responses is key to achieving desired outcomes. Additionally, consulting with a fitness professional can provide personalized guidance based on specific fitness goals and health considerations

Lesson 4. Personal Fitness Program

People participate in sports and other physically demanding pursuits because they value theirhealth and fitness. Some people will go to a gym where they will work with personal trainers (PT) who will assist them and design workout routines for them. However, it is possible that some people do not have the financial means to pay for a

PT; however, this does not preventthem from participating in physical activities. Learning how to design your fitness program will help you to continue working toward your goal of staying fit and healthy for the long term.

Training and strategy development methods for personal physical fitness activities will be the focus of this discussion. People who are working toward their goals of maintaining their fitness and health to take full advantage of life will find this to be helpful.

What is Personal Fitness Program (PFP)?

A personal exercise program is a plan that specifies the types and amounts of physical activity you will engage in to achieve your objectives. Programs are designed individually to meet the requirements of each participant. For instance, anyone looking

to lose weight or gain muscle mass will require a different strategy.

Importance of PFP

1. Prevents under or over-training

2. Gives Structure

3. Provides Goals

4. Provides a Checklist

5. Prevents Burnout

How to Design Your PFP?

1. **Consider Fitness Goal-** Are you going to begin a fitness program to assist you in losing weight? Or do you have another reason for wanting to get in shape, like training for a marathon? Having well-defined objectives can make it easier to evaluate your progress and keep you motivated.

2. **Assess Your fitness level.** - Assessment of your present fitness level will be your base benchmark to start your program design.

3. **Create a balanced routine** - It suggests that you spread out exercises during the course of a week.

Ex. Aerobic fitness

a. moderate Zumba activity for 150 minutes /week

b. vigorous rope jumping for 75 minutes/week.

c. Combination of moderate to vigorous for 150 minutes/week.

4. **Start low and progress slowly** - If you have never worked out before, you should ease into it and build up your endurance gradually.

Ex. Rope jumps 100 x/1st weeks and gradually increases the number of skips as you progress.

5. **Build activity into your daily routine** - Making time in your schedule for physical activity can be difficult. To make things simpler for yourself, put exercise on your calendar just as you would any other appointment.

Ex. Watching your favorite Netflix movie and doing your rope jump

6. **Plan to include different activities** - Doing constantly the same routine for a month will make you bored. Include different activities like rope jumps do jog next week. From low–moderate intensity, exercises move to high-intensity interval exercises with recovery.

7. **Allow time** - for recovery. Give yourself some time to recover. A lot of people start working out with overzealous enthusiasm, working out for too long or too intensely, and then they quit when their muscles and joints become sore or injured as a result of their workouts. Make sure you give your body enough time to recover and rest in between workouts.

Get Started*. Be committed to your plan*.

- Start slowly and build up gradually. Follow the phases of exercises.
- Break things up if you have to. If you feel you are already exhausted then break it out.
- Be creative. Find activities you enjoy to add to your fitness routine.
- Listen to your body. If you are exhausted or not feeling well, pains in doing the activity, then take a leave for your body to recover.

Monitor your progress-Perform a personal fitness assessment on yourself six weeks after you begin your training program, and then continue to do so every few months after. It is the realization that you need to extend the duration of your workouts to have the improvements you have made. If you are no longer motivated with your program change your goal.

Assessment Task 6

Part 1. Weekly Monitoring Plan

PERSONAL FITNESS PROGRAM TEMPLATE

Name:_____ Year&Section:_____

Weekly Exercise for the following days

Date:_____ to _____

Fitness Goal: _____

General Goal:_____

Specific Goal: _____

WEEK	DAY/DATE	EXERCISES	TIME	DURATION	HEART RATE	NOTES
1	1					
	2					
	ff					
	4					
	5					
	6					

COPY

Part 2: Practical Application (Performance)

1. **HIIT Workout Design:**

    - Design a HIIT workout plan for a specific fitness goal (e.g., cardiovascular endurance, fat loss, strength). Include details such as exercise selection, duration of work and rest intervals, and overall workout structure.

2. **Demonstration and Participation:**

    - Perform a HIIT workout based on the designed plan. This can be a recorded video or a live demonstration.
    - Emphasize proper form, intensity, and adherence to the planned intervals

Summary

In conclusion, High-Intensity Interval Training (HIIT) stands as a dynamic and transformative fitness approach that has reshaped the landscape of exercise. Its core principles of intensity, interval structure, and versatility have propelled it into the mainstream, appealing to individuals across various fitness levels and preferences

In essence, HIIT is not merely a workout routine; it represents a paradigm shift in how we approach fitness. Its adaptability, efficiency, and effectiveness make it a valuable tool for individuals striving to optimize their health and well-being. As we navigate the evolving landscape of fitness, HIIT stands as a beacon, offering a potent and accessible path towards achieving diverse fitness goals. Whether aiming for weight management, cardiovascular health, or overall fitness enhancement, HIIT emerges as a versatile and empowering solution, making a lasting impact on the way we perceive and engage in exercise.

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MODULE 6

CORE EXERCISES AND LOWER BODY EXERCISES

Introduction

The **core muscles** refer to a group of muscles located in the center of the body, including the abdominals, lower back, hips, and pelvis. These muscles are essential for maintaining **posture**, **balance**, and **stability**, and they play a critical role in nearly all physical activities, from walking and running to lifting and bending.

Core exercises are designed to strengthen these muscles, enhancing **core stability**, which is crucial for overall functional movement and injury prevention. Strong core muscles help improve posture, balance, and coordination, supporting the spine and pelvis and reducing the risk of lower back pain.

Intended Learning Outcomes

At the end of this module the students should be able to:

1. To identify and locate the different muscles of the core.
2. To discuss the importance of having more flexible and stronger core muscles.
3. To enumerate and demonstrate the core exercises

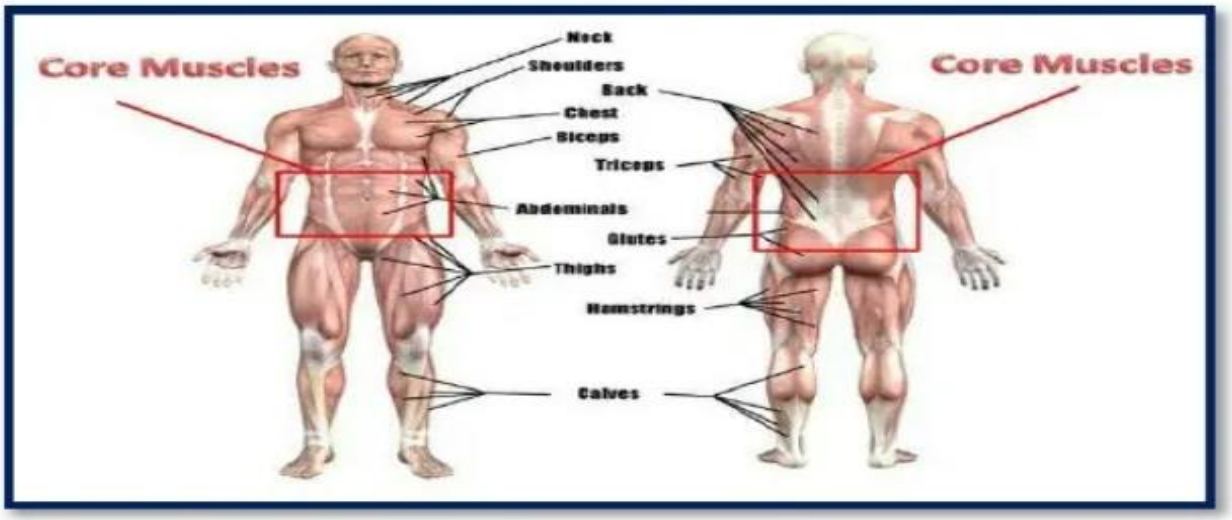
Lesson 1. Core Exercises

Core exercises are physical movements designed to strengthen the muscles in the core of your body. The "core" refers to the group of muscles in your abdomen, lower back, pelvis, and hips. These muscles are responsible for stabilizing your spine and pelvis, supporting your posture, and allowing for efficient movement in various physical activities.

The core is the **foundation** of most movements and is involved in almost every action we perform, whether it's walking, running, sitting, lifting, or playing sports. Having a strong core is essential for maintaining proper posture, balance, and stability, as well as preventing injuries, especially in the lower back and neck.

Core Muscles Include:

- **Abdominals**: Muscles in the front of the abdomen (e.g., rectus abdominis, transverse abdominis).
- **Obliques**: Muscles on the sides of the abdomen, responsible for rotation and side bending.
- **Lower Back Muscles**: Muscles that support the spine (e.g., erector spinae).
- **Glutes**: Muscles in the buttocks that stabilize the pelvis and hips.
- **Hip Flexors**: Muscles that allow you to bend the hip and stabilize the pelvis.
- **Pelvic Floor Muscles**: Muscles that support the organs in the pelvic region.



Recommended HRF every session

Muscular strength

Two or three 20-minute sessions each week that include exercises for all the major muscle groups are required. Lifting weights is one of the most effective ways to increase strength. For sedentary people, as little as two workouts per week can be beneficial.

Muscular endurance

Two to three 30-minute sessions each week that include exercises such as calisthenics, push-ups, curl-ups, pull-ups, and lightweight training for all the major muscle groups are required.

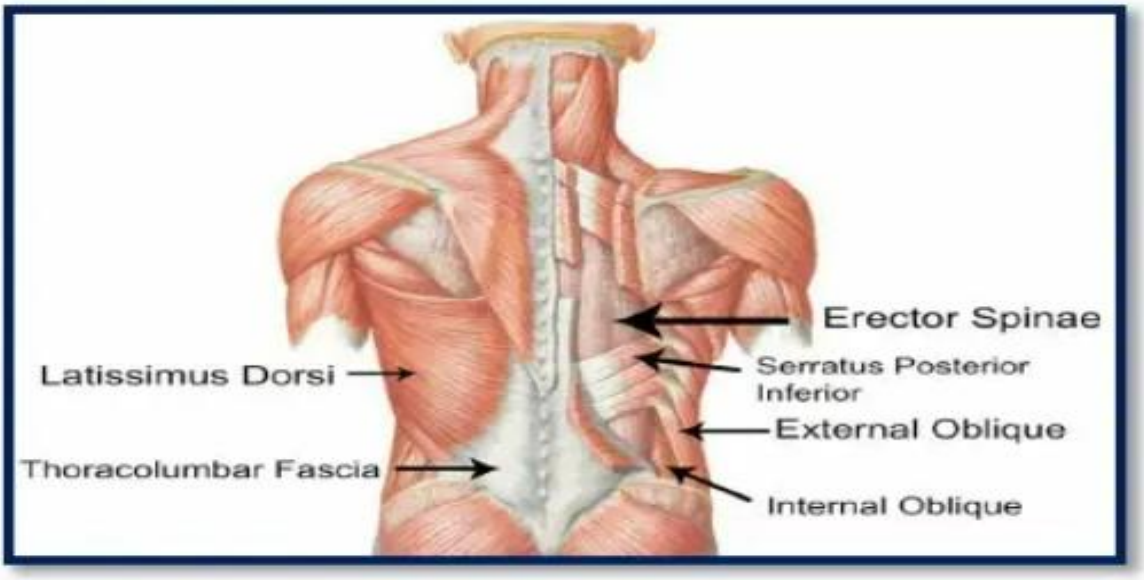
Cardiorespiratory endurance (CRE)

At least three 20- to 30-minute bouts of aerobic (activity requiring oxygen) exercise each week are recommended. Popular aerobic conditioning activities include brisk walking jogging swimming, cycling, rope jumping, rowing, cross-country skiing, and some continuous action games such as basketball and soccer.

Flexibility

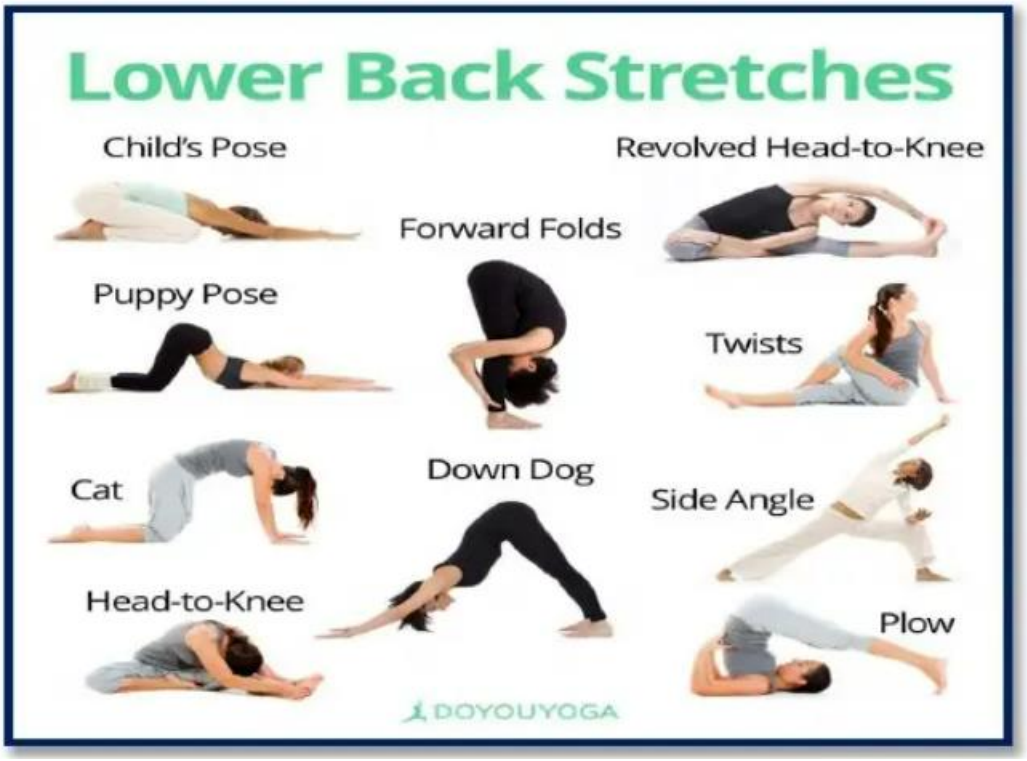
At the end of every workout, perform 5 to 10 minutes of static stretching exercises. If a workout session includes a CRE (Cardiorespiratory Endurance) session and a resistance training session, flexibility is best left to the end of the entire exercise routine

The **lower back muscles** are composed of Thoracolumbar Fascia, Erector Spinae, Serratus Posterior Inferior, and a portion of Latissimus Dorsi, External and Internal Obliques.(see picture below)



Tight and weak muscles can cause restricted mobility, body pains, injuries, and bad posture to name some consequences when a person does not do regular stretching and corestrengthening exercises. In contrast, that person will be able to reach upward or be nddownward, forward or sideways gracefully when he does stretch regularly. Moreover, the lower back muscles become stronger and are able to maintain a certain body position. Postures noticeably good when the core muscles are trained well.

What, then, are the core stretches and exercises that you can do?



As you do the stretches, make sure to keep on breathing in a relaxed manner. This is to enable the exchange of gases in your lungs and in the muscles for optimal health benefits. Please note from the photo above that the 'Forward Folds' can be done in a sitting position, which is safer

Forward Folds in sitting position:

You may begin the Forward Folds stretch with your upper body straight up and arms extended Infront of you at the sides of each thigh or on your lap.

Abdominal Stretch:

You may start with your elbows down on the floor.

Then, gradually, work your way up as you become more flexible in the abdominal area. Don't forget the regular, relaxed breathing.

BASIC CORE EXERCISES:

1. Crunches

2. Modified Plank – Knee Down

Plank – Legs Straight

3. Bicycles

Importance of Core Exercises:

1. **Posture Improvement:** A strong core helps to support proper posture, which can prevent slouching and lower back pain.

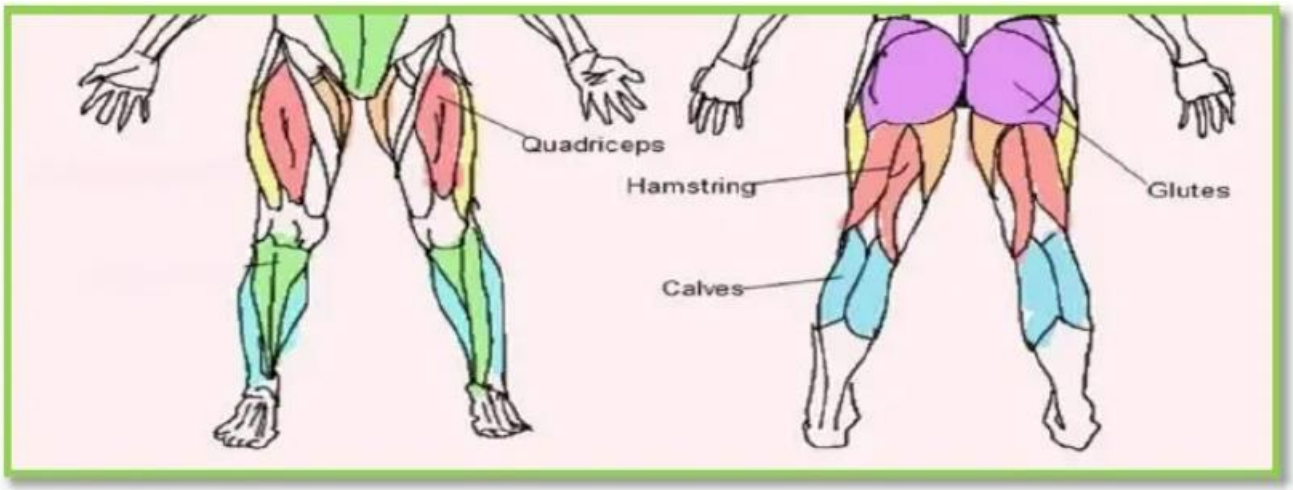
2. **Injury Prevention:** By strengthening the muscles that stabilize the spine and pelvis, core exercises reduce the risk of injuries, especially in the lower back.
3. **Balance and Stability:** A strong core improves your ability to maintain balance, whether during physical activity or daily movements.
4. **Enhanced Athletic Performance:** A well-developed core is crucial for nearly all sports and physical activities, as it provides a stable foundation for generating power and movement.
5. **Functional Movement:** Core strength is necessary for a wide range of daily activities, such as lifting, bending, twisting, and reaching.
6. **Pain Reduction:** Strengthening the core can help alleviate pain, particularly in the lower back, by improving posture and supporting the spine.

Lesson 2. Lower Body Exercises

Lower body exercises are movements designed to strengthen the muscles in the legs, hips, and glutes (buttocks). These exercises target key muscle groups such as the quadriceps, hamstrings, gluteus maximus, calves, and hip flexors, which are essential for movement and stability in the lower half of the body.

Strengthening the lower body is crucial for improving overall physical fitness, athletic performance, and functional movement in daily life. Lower body exercises help you perform essential tasks such as walking, running, lifting, squatting, and climbing stairs, while also playing a major role in posture, balance, and mobility.

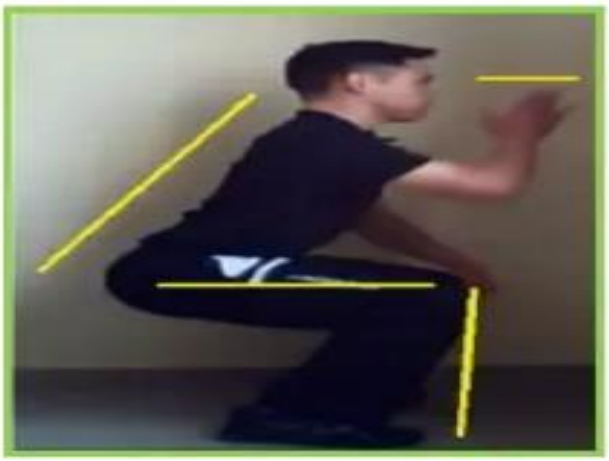
Lower body muscles are the muscles found in our buttocks, thighs, and legs. The names of these common muscles are the gluteal muscles (‘glutes’), quadriceps femoris(‘quads’), hamstring muscles, and the gastrocnemius muscles (‘calves’)



Having more flexible and stronger muscles in the lower body prevents injuries, backpain, and poor posture. The body is able to reach farther or higher. It can climb up the stairs with ease. It exert more power when the legs straighten up such as the kicking motion or jumping action. When the leg muscles are strong, it supports and protects the knees from serious damage. Man can even walk for many more years in his lifetime

Some of the **LOWER BODY EXERCISES** that are functional in nature are:

1. Squats – As we sit down and rise up from the chair, as you lift a pail or two of water, as youlift heavy objects such as tables or house furniture.



Observe the alignment of your body parts. Inhale as you go down, exhale as you stand up.

2. **Side Lunges –** As you pick up your balance, as you return a ball in a racket sport, as you try to reach in a sideways position.

3. **Squat and Press Up** – As you lift, carry, and place an object on a cabinet.

4. **High Knees –** An aerobic exercise to improve the capability of your heart and lungs. This exercise is recommended for people with health.

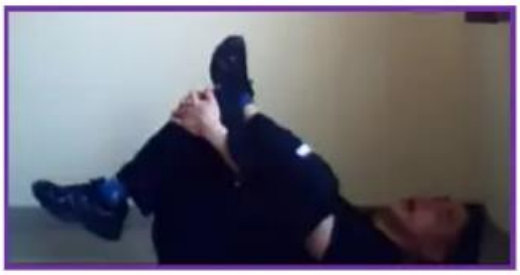
5. **Lunges –** As you recover your balance from an unstable position, this exercise strengthens your thigh muscles and protects the knees from harm.

LOWER BODY STRETCHES

1. Quadriceps Stretch

2. Hamstring Stretch

3. Glute Stretch – pull your knee towards your chest. Again, don't forget to breathe.



4. Calf Stretch – hands on the wall, one foot flat backward and that same leg extended, and feel it where I am pointing at.

GUIDELINE IN EXERCISING

1. Consult with your physician or family doctor. You need to know if you are good to engage in physical activities or exercises.

2. You need to find out what exercises are allowed and prohibited.

3.. Wear appropriate clothing and rubber shoes for safety and for free-flowing movements.

4. Do not come on an empty stomach. Your body needs energy as fuel for the entire session.

5. When exercising alone, do it in a hazard-free environment or place for your safety.

Importance of Lower Body Exercises:

1. **Increased Strength and Power**: Lower body exercises help build the strength needed to perform various physical activities, whether it's lifting, jumping, or sprinting. Strong legs and hips contribute to overall physical power.
2. **Improved Balance and Stability**: A strong lower body enhances balance and stability, reducing the risk of falls and injuries, especially as we age.
3. **Enhanced Athletic Performance**: Lower body strength is critical in nearly all sports, including running, soccer, basketball, cycling, and skiing. A strong lower body improves speed, endurance, agility, and explosiveness.
4. **Functional Movement**: Lower body muscles are heavily involved in day-to-day activities such as walking, sitting, standing, and bending. Strengthening these muscles can make these movements easier and more efficient.
5. **Injury Prevention**: Strong lower body muscles support the knees, hips, and ankles, reducing the risk of common injuries like strains, sprains, and ligament tears. These exercises also improve joint health by strengthening the muscles around the joints.
6. **Boosted Metabolism**: Because the lower body contains large muscle groups, strengthening these muscles increases metabolism, which helps burn more calories even at rest.

Key Lower Body Muscles:

1. **Quadriceps (Quads)**: Located at the front of the thigh, the quadriceps are responsible for knee extension and play a key role in walking, running, and jumping.
2. **Hamstrings**: Found at the back of the thigh, the hamstrings are responsible for bending the knee and extending the hip.
3. **Glutes**: The gluteus maximus, gluteus medius, and gluteus minimus make up the muscles of the

buttocks, which are involved in hip extension, stability, and rotation.

4. **Calves**: The calf muscles (gastrocnemius and soleus) are responsible for ankle movement and help with walking, running, and jumping.
5. **Hip Flexors**: These muscles (including the iliopsoas) are responsible for lifting the thigh and bending the hip.
6. **Adductors and Abductors**: The adductors are responsible for bringing the legs together (adduction), while the abductors (such as the gluteus medius) help to move the legs outward (abduction).

Assessment Task 7

Instructions: For this part of the assessment, perform the following exercises and demonstrate your ability to correctly execute each movement. You will be assessed on form, technique, and the ability to maintain proper body alignment. You may be required to submit a video or perform the exercises during a scheduled assessment session.

1. **Core Exercise Routine:**
    - **Plank** (Hold for 30 seconds to 1 minute)
    - **Russian Twists** (10-20 reps per side)
    - **Dead Bugs** (10-15 reps per side)
    - **Superman Exercise** (Hold for 30 seconds to 1 minute)

1. Plank

2. Russian Twists

3. Dead Bug

4. Superman Stretch

Assessment Criteria:

- Proper body alignment during the plank.
- Engaging the core during Russian twists without excessive twisting of the lower back.
- Maintaining a neutral spine in dead bugs and Superman exercises.
- Consistent movement and proper technique throughout.

2. **Lower Body Exercise Routine:**

- **Squats** (10-15 reps)
- **Lunges** (10-12 reps per leg)
- **Glute Bridges** (10-15 reps)
- **Calf Raises** (15-20 reps)

1. Squat

2. Lunges

3. Glute Bridges

4. Calf Raises

Assessment Criteria:

- Correct squat depth with proper knee and hip alignment.
- Maintaining balance and proper knee alignment during lunges.
- Engaging the glutes and core during glute bridges.
- Performing calf raises with full ankle range of motion.

3. Combination Exercise (Core and Lower Body):

- Mountain Climbers (15-20 reps per leg)
- Kettlebell Swings (10-15 reps)
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1. Mountain Climbers

2. Kettlebell Swings

Assessment Criteria:

- Performing mountain climbers with controlled movement and core engagement.
- Correct kettlebell swing technique, using the hips to generate power while maintaining a neutral spine.

Summary

Both core and lower body exercises are essential for developing functional fitness. The core provides stability and support for lower body movements, while strong lower body muscles allow for more powerful and efficient movement. Together, these exercises improve posture, balance, mobility, and athletic performance.

By incorporating both types of exercises into a fitness routine, individuals can experience better overall strength, reduce the risk of injury, and enhance both daily activities and sports performance.

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